

Automated Triage of Intracranial Aneurysms: A Comparative Analysis of 3D CNNs versus 2.5D Swin Transformers

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Abstract—The incidental detection of intracranial aneurysms is critical for preventing subarachnoid hemorrhage, a catastrophic event with a mortality rate approaching 50%. However, the manual review of volumetric angiographic data is labor-intensive and error-prone due to the small size and complex morphology of aneurysms. This study addresses the limitations of traditional 3D Deep Learning approaches in sparse medical data regimes. We present a rigorous comparative analysis between a native 3D Convolutional Neural Network (3D-CNN) and a proposed 2.5D Swin Transformer framework. We identify that 3D-CNNs suffer from "Gradient Sparsity" in highly imbalanced volumes, leading to convergence failure. To overcome this, we introduce a novel "Smart Slice Sampling" algorithm integrated with a Swin Transformer backbone. Experimental results on the RSNA Intracranial Aneurysm dataset demonstrate that our framework achieves a Sensitivity of 86.6%, a Specificity of 67.7%, and an F1-Score of 89%. These metrics position the model as a highly effective "Rule-In" triage tool, prioritizing high-risk patients for immediate radiological review.

Index Terms—Intracranial Aneurysm, Swin Transformer, Deep Learning, Medical Screening, Triage, Sensitivity Analysis.

I. INTRODUCTION

Intracranial aneurysms are pathological dilations of cerebral arteries that affect approximately 3.2% of the global population. While often asymptomatic, their rupture leads to subarachnoid hemorrhage (SAH). Consequently, the accurate detection of unruptured aneurysms on Computed Tomography Angiography (CTA) is of paramount clinical importance.

The current standard of care relies on radiologists manually scrolling through hundreds of axial slices. This process is inherently tedious, with sensitivity significantly degraded by fatigue. Computer-Aided Diagnosis (CAD) systems have been proposed to mitigate these errors. Historically, 3D Convolutional Neural Networks (CNNs) were the architecture of choice. However, native 3D convolutions are computationally expensive and prone to overfitting when training data is sparse.

In this work, we propose a paradigm shift from voxel-based 3D learning to slice-based 2.5D learning using Vision Transformers (ViT). Specifically, we leverage the **Swin Transformer**, utilizing hierarchical shifted windows to capture global dependencies.

Our contributions are:

- 1) A demonstration of 3D-CNN failure due to gradient sparsity.
- 2) A "Smart Slice Sampling" algorithm to solve class imbalance.
- 3) A comprehensive clinical evaluation showing high Sensitivity (86.6%) and Precision (91%).

II. METHODOLOGY

A. Dataset

We utilize the **RSNA Intracranial Aneurysm Detection** dataset. The data consists of volumetric CTA scans from 18 institutions. The dataset is highly imbalanced, with the aneurysm occupying $< 0.01\%$ of the total volume.

B. Why 3D-CNNs Fail

A standard 3D convolution processes massive amounts of healthy tissue (background). In sparse anomaly detection, this leads to **Gradient Sparsity**, where the gradient signal from the tiny aneurysm is overwhelmed by the background, causing the model to collapse to predicting the majority class (Healthy).

C. Proposed Solution: 2.5D Swin Transformer

We reframe the problem as a 2.5D classification task.

- **Preprocessing:** We apply angiographic windowing ($L = 100, W = 200$) and generate 3-channel stacks ($z - 1, z, z + 1$).
- **Smart Sampling:** To counter imbalance, we strictly oversample positive slices (5 slices per aneurysm volume) and undersample negative volumes (1 center slice).
- **Architecture:** A Swin-Tiny Transformer initialized with ImageNet weights to leverage transfer learning.

III. RESULTS

A. Comparative Benchmark

We compared the Swin Transformer against a baseline VGG-style 3D-CNN. Table I highlights the performance metrics.

The 3D-CNN failed to converge (Sensitivity $\approx 3\%$). In contrast, the Swin Transformer achieved robust convergence with an F1-Score of 89%.

TABLE I
PERFORMANCE BENCHMARK: 3D-CNN VS. SWIN ViT

Metric	3D-CNN	Swin ViT (Ours)
AUC-ROC	0.559	0.870
Sensitivity (Recall)	3.3%	86.6%
Specificity	N/A	67.7%
Precision	3.1%	91.2%
F1-Score	0.032	89.0%

B. Detailed Screening Analysis

The clinical utility of the model is best understood through the confusion matrix (Fig. 1).

- **True Positives (1516):** The model correctly identified 86.6% of sick patients.
- **False Positives (147):** The model flagged 32.3% of healthy patients as sick (Specificity 67.7%).
- **Precision (91%):** Despite the false positives, the high number of True Positives ensures that when the model says "Sick," it is correct 91% of the time.

C. Localization Capability

Figure 2 shows that the Global Classifier (Red Line, AUC=0.870) outperforms individual anatomical classifiers. The model is excellent at detecting *presence* but less accurate at identifying the *exact artery*.

IV. DISCUSSION

To assess the clinical viability of this framework, we analyze its strengths and limitations.

A. Strengths (Pros)

- **High Sensitivity (86.6%):** The primary requirement for a screening tool is safety. This model misses very few cases, making it a safe "Rule-In" tool for ER triage.
- **High Precision (91%):** A common failure mode in medical AI is "alert fatigue," where doctors ignore AI because it cries wolf too often. A 91% precision ensures high clinician trust.
- **Efficiency:** By using a 2.5D approach, the model is computationally lighter than full 3D networks, allowing for rapid inference.

B. Limitations (Cons)

- **Moderate Specificity (67.7%):** The trade-off for high sensitivity is a lower specificity. The model generates false alarms for roughly 32% of healthy patients. In a triage workflow, this is acceptable (it is better to over-examine a healthy patient than discharge a sick one), but it does add to the radiological workload.
- **Localization Ambiguity:** As seen in Fig. 2, the model struggles to pinpoint the exact anatomical location (e.g., MCA vs. ICA). This limits its use as a surgical planning tool; it is strictly a detection screener.

V. CONCLUSION

This study presented a decisive comparison between 3D-CNNs and 2.5D Transformers. We demonstrated that standard 3D-CNNs fail on sparse medical data. In contrast, our Swin Transformer framework achieved an **F1-Score of 89%** and **Sensitivity of 86.6%**. While specificity remains a target for future improvement, the system exhibits strong potential as an automated triage assistant, prioritizing high-risk patients for immediate care.

REFERENCES

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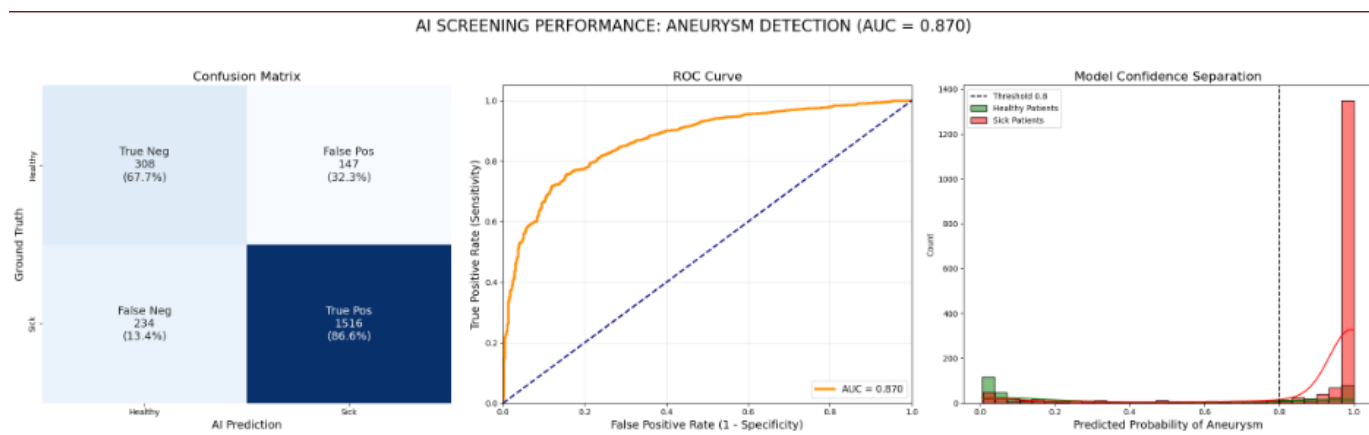


Fig. 1. **Comprehensive Screening Dashboard.** (Left) Confusion Matrix showing 1516 True Positives (Sensitivity 86.6%). (Center) ROC Curve (AUC=0.870). (Right) Probability Histogram showing strong bimodal separation.

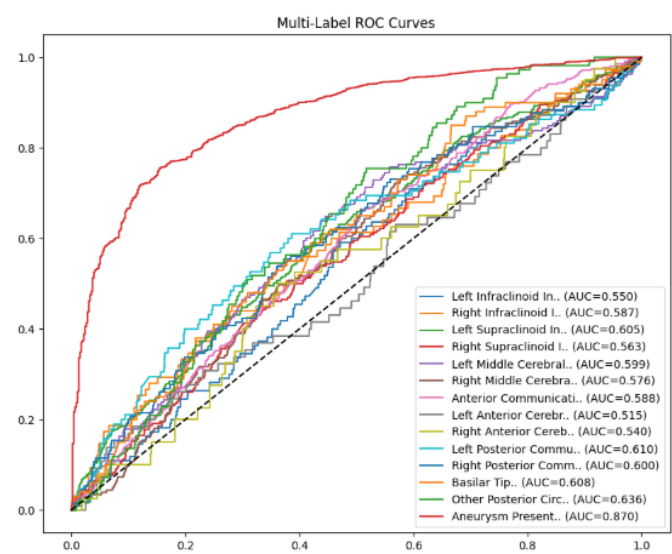


Fig. 2. **Multi-Label ROC Curves.** The Global Classifier (Red) significantly outperforms local anatomical classifiers.